Physics I

Code : EBA1110

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Course Content

- Lecture 1: Units and Vectors.
- Lecture 2: Newton Laws of motion &Gravitational Force, weight and mass
- Lecture 3: Work, Kinetic and Potential Energy, Work Energy theorem& Conservation law of Energy

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- Lecture 4: Oscillatory motion , simple and Harmonic Motion, Introduction to mechanical waves properties , and Travelling wave
- Lecture 5: Sound wave Power, Intensity, speed in solid , liquid or gas , Sound level calculation
- Lecture 6: Superposition of waves.

Course Content

- Lecture 7: capacitors.
- Lecture 8: Calculation of the Electric Current, Resistance, Ohm's law, Power Calculation.
- Lecture 9: Electromotive force , Internal Resistance , Resistance in Series and Parallel, Kirchhoff law and RC Circuit
- Lecture 10: Magnetic Force
- Lecture 11: Biot Savart Law, Ampere Law and their applications. Magnetic force for two parallel conductors.Explanation of Faraday Law of Induction

LECTURE 3

- Work Definition
- Work due to constant and variable force
- Kinetic and Potential Energy
- Work Energy theorem
- Conservation law of Energy

1-Work Done by a Constant Force

Let us examine the situation in Figure 7.2, where the object (the system) undergoes a displacement along a straight line while acted on by a constant force of magnitude F that makes an angle Θ with the direction of the displacement

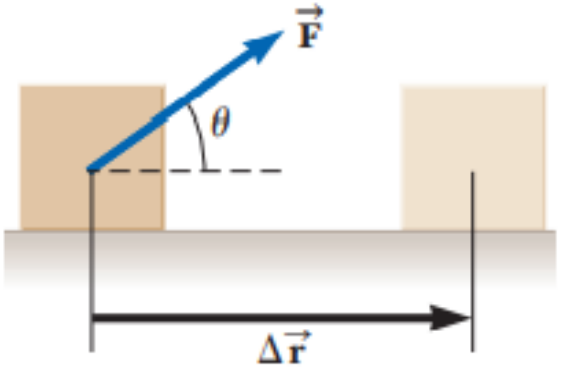


Figure 7.2 An object undergoes a displacement $\Delta\vec{\mathbf{r}}$ under the action of a constant force $\vec{\mathbf{F}}$.

1 -Work Definition :

The **work** W done on a system by an agent exerting a constant force on the system is the product of the magnitude F of the force, the magnitude Δr of the displacement of the point of application of the force, and $\cos\theta$, where θ is the angle between the force and displacement vectors:

$$W \equiv F\,\Delta r \cos\theta \qquad (7.1)$$

Notice in Equation 7.1 that work is a scalar, even though it is defined in terms of two vectors, a force $\vec{\mathbf{F}}$ and a displacement $\Delta\vec{\mathbf{r}}$.

Notice also that the displacement in Equation 7.1 is that of *the point of application of the force*. If the force is applied to a particle or a rigid object that can be modeled as a particle, this displacement is the same as that of the particle. For a deformable system, however, these displacements are not the same.

Also notice from Equation 7.1 that the work done by a force on a moving object is zero when the force applied is perpendicular to the displacement of its point of application. That is, if $\theta = 90°$, then $W = 0$ because $\cos 90° = 0$.

The sign of the work also depends on the direction of $\vec{\mathbf{F}}$ relative to $\Delta\vec{\mathbf{r}}$. The work done by the applied force on a system is positive when the projection of $\vec{\mathbf{F}}$ onto $\Delta\vec{\mathbf{r}}$ is in the same direction as the displacement. For example, when an object is lifted, the work done by the applied force on the object is positive because the direction of that force is upward, in the same direction as the displacement of its point of application. When the projection of $\vec{\mathbf{F}}$ onto $\Delta\vec{\mathbf{r}}$ is in the direction opposite the displacement, W is negative.

If an applied force $\vec{\mathbf{F}}$ is in the same direction as the displacement $\Delta\vec{\mathbf{r}}$, then $\theta = 0$ and $\cos 0 = 1$. In this case, Equation 7.1 gives

$$W = F\,\Delta r$$

The units of work are those of force multiplied by those of length. Therefore, the SI unit of work is the **newton · meter** ($\mathrm{N \cdot m = kg \cdot m^2/s^2}$). This combination of units is used so frequently that it has been given a name of its own, the **joule** (J).

The scalar product of any two vectors $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$ is defined as a scalar quantity equal to the product of the magnitudes of the two vectors and the cosine of the angle θ between them:

$$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} \equiv AB\cos\theta \tag{7.2}$$

As is the case with any multiplication, $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$ need not have the same units.

By comparing this definition with Equation 7.1, we can express Equation 7.1 as a scalar product:

$$W = F\,\Delta r\cos\theta = \vec{\mathbf{F}} \cdot \Delta\vec{\mathbf{r}} \tag{7.3}$$

In other words, $\vec{\mathbf{F}} \cdot \Delta\vec{\mathbf{r}}$ is a shorthand notation for $F\,\Delta r\cos\theta$.

2-Work Done by a Varying Force

Consider a particle being displaced along the x axis under the action of a force that varies with position. In such a situation, we cannot use Equation 7.1 to calculate the work done by the force.

- Figure 7.7a shows a varying force applied on a particle that moves from initial position X_i to final position X_f .
- Imagine a particle undergoing a very small displacement Δx, shown in the figure.
- The x component of the force (F_x) is approximately constant over this small interval;
- so for this small displacement, we can approximate the work done on the particle by the force using
- Equation 7.1 as:

$$W \approx F_x \Delta x$$

- which is the area of the shaded rectangle in Figure 7.7a.

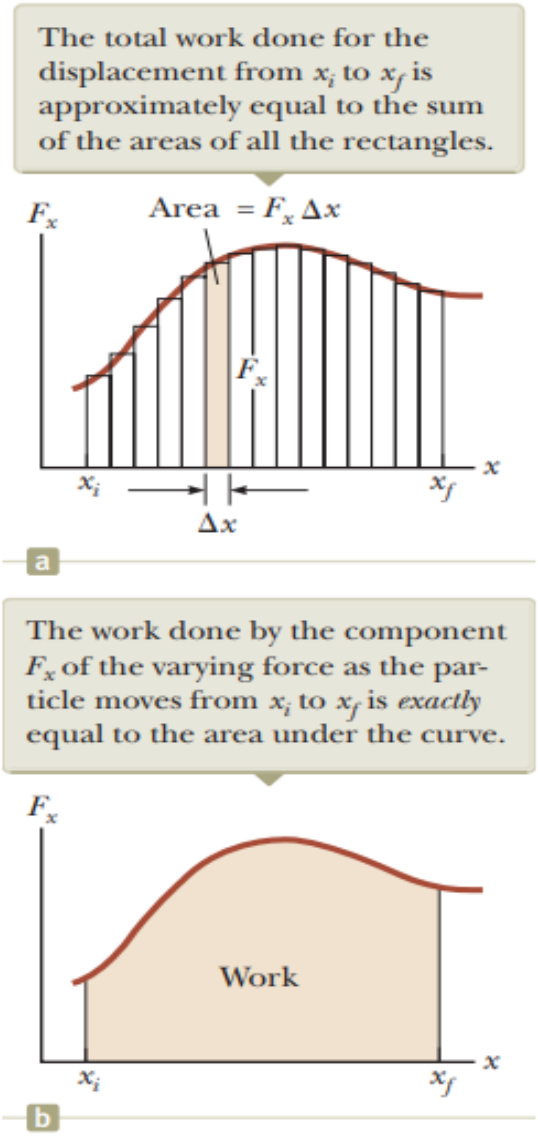


Figure 7.7 (a) The work done on a particle by the force component F_x for the small displacement Δx is $F_x \Delta x$, which equals the area of the shaded rectangle. (b) The width Δx of each rectangle is shrunk to zero.

If the Fx versus x curve is divided into a large number of such intervals, the total work done for the displacement from X_i to X_f is approximately equal to the sum of a large number of such terms

$$W \approx \sum_{x_i}^{x_f} F_x \, \Delta x$$

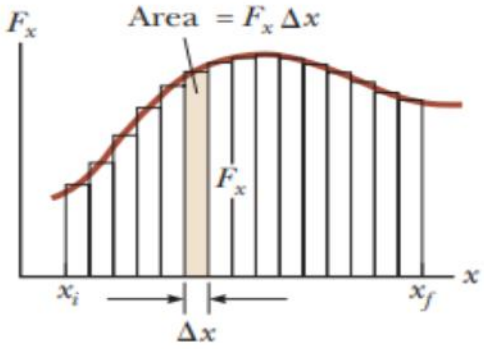
If the size of the small displacements is allowed to approach zero ($\Delta x \rightarrow 0$),
the number of terms in the sum increases without limit but the value of the sum approaches a definite value equal to the area bounded by the Fx curve and the x axis
And from the definision of the integration proces which equal to the area bounded by the curve

$$\lim_{\Delta x \to 0} \sum_{x_i}^{x_f} F_x \, \Delta x = \int_{x_i}^{x_f} F_x \, dx$$

Therefore, we can express the work done by Fx on the system of the particle as it moves from xi to xf as

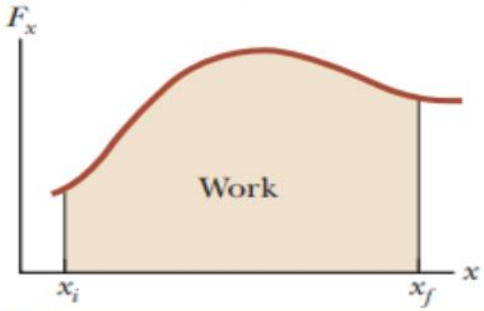
$$W = \int_{x_i}^{x_f} F_x \, dx \qquad (7.7)$$

The total work done for the displacement from x_i to x_f is approximately equal to the sum of the areas of all the rectangles.



a

The work done by the component F_x of the varying force as the particle moves from x_i to x_f is *exactly* equal to the area under the curve.



b

Figure 7.7 (a) The work done on a particle by the force component F_x for the small displacement Δx is $F_x \Delta x$, which equals the area of the shaded rectangle. (b) The width Δx of each rectangle is shrunk to zero.

- If more than one force acts on a system and the system can be modeled as a particle,
  the total work done on the system is just the work done by the net force.
- If we express the net force in the x direction as ΣF_x,
  the total work, or net work, done as the particle moves from x_i to x_f is

$$\sum W = W_{\text{ext}} = \int_{x_i}^{x_f} \left(\sum F_x\right) dx \quad \text{(particle)}$$

For the general case of a net force $\sum \vec{\mathbf{F}}$ whose magnitude and direction may both vary, we use the scalar product,

$$\sum W = W_{\text{ext}} = \int \left(\sum \vec{\mathbf{F}}\right) \cdot d\vec{\mathbf{r}} \quad \text{(particle)} \qquad (7.8)$$

The subscript "ext" on work reminds us that the net work is done by an external agent on the system

If the system cannot be modeled as a particle (for example, if the system is deformable), we cannot use Equation 7.8 because different forces on the system may move through different displacements. In this case, we must evaluate the work done by each force separately and then add the works algebraically to find the net work done on the system:

$$\sum W = W_{\text{ext}} = \sum_{\text{forces}} \left(\int \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \right) \quad \text{(deformable system)}$$

Example 7.4 Calculating Total Work Done from a Graph

A force acting on a particle varies with x as shown in Figure 7.8. Calculate the work done by the force on the particle as it moves from $x = 0$ to $x = 6.0$ m.

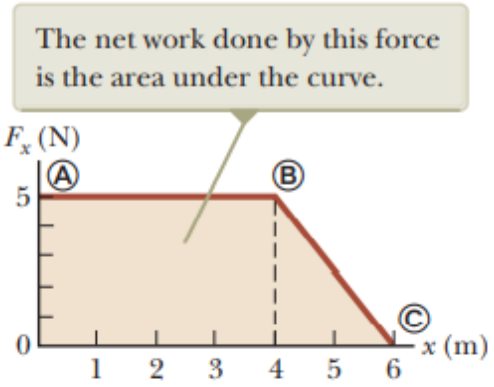


Figure 7.8 (Example 7.4) The force acting on a particle is constant for the first 4.0 m of motion and then decreases linearly with x from $x_{Ⓑ} = 4.0$ m to $x_{Ⓒ} = 6.0$ m.

SOLUTION

Conceptualize Imagine a particle subject to the force in Figure 7.8. The force remains constant as the particle moves through the first 4.0 m and then decreases linearly to zero at 6.0 m. In terms of earlier discussions of motion, the particle could be modeled as a particle under constant acceleration for the first 4.0 m because the force is constant. Between 4.0 m and 6.0 m, however, the motion does not fit into one of our earlier analysis models because the acceleration of the particle is changing. If the particle starts from rest, its speed increases throughout the motion, and the particle is always moving in the positive x direction. These details about its speed and direction are not necessary for the calculation of the work done, however.

Categorize Because the force varies during the motion of the particle, we must use the techniques for work done by varying forces. In this case, the graphical representation in Figure 7.8 can be used to evaluate the work done.

Analyze The work done by the force is equal to the area under the curve from $x_{Ⓐ} = 0$ to $x_{Ⓒ} = 6.0$ m. This area is equal to the area of the rectangular section from Ⓐ to Ⓑ plus the area of the triangular section from Ⓑ to Ⓒ.

Evaluate the area of the rectangle: $W_{Ⓐ \text{ to } Ⓑ} = (5.0 \text{ N})(4.0 \text{ m}) = 20 \text{ J}$

Evaluate the area of the triangle: $W_{Ⓑ \text{ to } Ⓒ} = \frac{1}{2}(5.0 \text{ N})(2.0 \text{ m}) = 5.0 \text{ J}$

Find the total work done by the force on the particle: $W_{Ⓐ \text{ to } Ⓒ} = W_{Ⓐ \text{ to } Ⓑ} + W_{Ⓑ \text{ to } Ⓒ} = 20 \text{ J} + 5.0 \text{ J} = \boxed{25 \text{ J}}$

Finalize Because the graph of the force consists of straight lines, we can use rules for finding the areas of simple geometric models to evaluate the total work done in this example. If a force does not vary linearly as in Figure 7.7, such rules cannot be used and the force function must be integrated as in Equation 7.7 or 7.8.

Kinetic Energy and the Work-Kinetic Energy Theorem

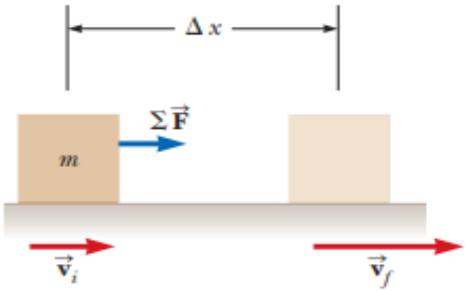


Figure 7.12 An object undergoing a displacement $\Delta\vec{r} = \Delta x\hat{i}$ and a change in velocity under the action of a constant net force $\Sigma\vec{F}$.

In this section, we investigate this situation and introduce our first type of energy that a system can possess, called *kinetic energy*.

Consider a system consisting of a single object. Figure 7.12 shows a block of mass m moving through a displacement directed to the right under the action of a net force $\Sigma\vec{F}$, also directed to the right. We know from Newton's second law that the block moves with an acceleration $\vec{a}$. If the block (and therefore the force) moves through a displacement $\Delta\vec{r} = \Delta x\hat{i} = (x_f - x_i)\hat{i}$, the net work done on the block by the external net force $\Sigma\vec{F}$ is

$$W_{ext} = \int_{x_i}^{x_f} \Sigma F\, dx \qquad (7.14)$$

Using Newton's second law, we substitute for the magnitude of the net force $\Sigma F = ma$ and then perform the following chain-rule manipulations on the integrand:

$$W_{ext} = \int_{x_i}^{x_f} ma\, dx = \int_{x_i}^{x_f} m\frac{dv}{dt}\, dx = \int_{x_i}^{x_f} m\frac{dv}{dx}\frac{dx}{dt}\, dx = \int_{v_i}^{v_f} mv\, dv$$

$$W_{ext} = \tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_i^2 \qquad (7.15)$$

where v_i is the speed of the block at $x = x_i$ and v_f is its speed at x_f.

Equation 7.15 was generated for the specific situation of one-dimensional motion, but it is a general result. It tells us that the work done by the net force on a particle of mass m is equal to the difference between the initial and final values of a quantity $\frac{1}{2}mv^2$. This quantity is so important that it has been given a special name, **kinetic energy:**

$$K \equiv \tfrac{1}{2}mv^2 \qquad (7.16)$$

Kinetic energy represents the energy associated with the motion of the particle.

Note that kinetic energy is a scalar quantity and has the same units as work.

It is often convenient to write Equation 7.15 in the form

$$W_{ext} = K_f - K_i = \Delta K \qquad (7.17)$$

work–kinetic energy theorem:

Equation 7.17 is an important result known as the **work–kinetic energy theorem:**

When work is done on a system and the only change in the system is in its speed, the net work done on the system equals the change in kinetic energy of the system, as expressed by Equation 7.17:

Potential Energy of a System

we can identify the quantity mgy as the gravitational potential energy Ug of the system of an object of mass m and the Earth:

$$U_g = mgy \qquad (7.19)$$

The units of gravitational potential energy are joules, the same as the units of work and kinetic energy. Potential energy, like work and kinetic energy, is a scalar quantity. Notice that Equation 7.19 is valid only for objects near the surface of the Earth, where g is approximately constant.[3]

Consider an external agent lifting an object of mass m from an initial height y_i above the ground to a final height y_f as in Figure 7.15. We assume the lifting is done slowly, with no acceleration, so the applied force from the agent is equal in magnitude to the gravitational force on the object: the object is modeled as a particle in equilibrium moving at constant velocity. The work done by the external agent on the system (object and the Earth) as the object undergoes this upward displacement is given by the product of the upward applied force $\vec{\mathbf{F}}_{app}$ and the upward displacement of this force, $\Delta\vec{\mathbf{r}} = \Delta y\hat{\mathbf{j}}$:

$$W_{ext} = (\vec{\mathbf{F}}_{app}) \cdot \Delta\vec{\mathbf{r}} = (mg\hat{\mathbf{j}}) \cdot [(y_f - y_i)\hat{\mathbf{j}}] = mgy_f - mgy_i$$

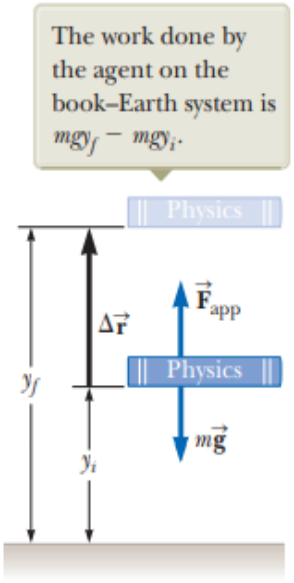


Figure 7.15 An external agent lifts a book slowly from a height y_i to a height y_f.

Using our definition of gravitational potential energy,

$$W_{ext} = \Delta U_g$$

which mathematically describes that the net external work done on the system in this situation appears as a change in the gravitational potential energy of the system.

Gravitational potential energy depends only on the vertical height of the object above the surface of the Earth.

Conservation law of Energy

The general statement of the principle of conservation of energy can be described mathematically with the conservation of energy equation as follows:

$$\Delta E_{\text{system}} = \sum T \qquad (8.1)$$

where E_{system} is the total energy of the system, including all methods of energy storage (kinetic, potential, and internal), and T (for *transfer*) is the amount of energy transferred across the system boundary by some mechanism. Two of our transfer mechanisms have well-established symbolic notations. For work, $T_{\text{work}} = W$ as discussed in Chapter 7, and for heat, $T_{\text{heat}} = Q$ as defined in Chapter 20. (Now that we are familiar with work, we can simplify the appearance of equations by letting the simple symbol W represent the external work W_{ext} on a system. For internal work, we will *always* use W_{int} to differentiate it from W.) The other four members of our list do not have established symbols, so we will call them T_{MW} (mechanical waves), T_{MT} (matter transfer), T_{ET} (electrical transmission), and T_{ER} (electromagnetic radiation).

The full expansion of Equation 8.1 is

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W + Q + T_{\text{MW}} + T_{\text{MT}} + T_{\text{ET}} + T_{\text{ER}} \qquad (8.2)$$

The change in the total amount of energy in the system is equal to the total amount of energy that crosses the boundary of the system.